



Research article

Artificial intelligence and academic integrity in nursing education: A mixed methods study on usage, perceptions, and institutional implications

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ARTICLE INFO

Keywords:

Academic integrity
Academic misconduct
Artificial intelligence
Implications
Intention to adopt
Nursing education
Nursing students
Mixed methods

ABSTRACT

Background: The rise of artificial intelligence (AI) use in higher education has generated substantial debate among academics and students, given the potential for students to engage in academic misconduct through the misuse of AI. Academics argue that AI poses a serious threat to the foundational development of nurses through the questionable integrity of AI-generated academic work and by undermining the development of critical thinking skills essential for professional practice. However, there is limited research on nursing students' integration of AI technologies in their studies.

Method: This study utilised a convergent parallel mixed methods approach to develop a multiphase approach with convergent parallel techniques for the qualitative and quantitative phases. The quantitative method utilised a Qualtrics-powered online survey to engage 188 nursing students, exploring various domains related to AI use. In the qualitative phase, in-depth interviews with 13 purposively sampled students provided deeper insights. The qualitative data were analysed using an inductive thematic analysis approach, while the quantitative data were analysed using SPSS.

Result: In the survey, 24 % of respondents reported using AI, ranging from moderate to extensive usage. In logistics regression analysis, hearing about AI (OR = 3.9; CI 1.07–10.2; $p < 0.05$), the belief that AI was useful in the studies (OR = 5.5; CI 1.7–17.3; $p < 0.01$), and the perception that learning to use AI is easy (OR = 3.4; CI 1.1–11.1; $p < 0.05$) predicted AI use. Qualitative findings revealed that all students used AI for various academic purposes. The 'fascinating', 'intelligent' and 'efficient' nature of AI in handling 'time-consuming' academic tasks motivated its use. However, concerns about breaching academic integrity and the value of achieving success through personal effort served as deterrents.

Conclusion: The findings suggest that while AI's efficiency drives students to adopt it, they remain cautious about its ethical implications, leading to uncertainty in its application within academic practices. This highlights the critical need for institutional support and explicit guidelines on responsible AI integration in educational settings.

1. Introduction

Artificial intelligence (AI) in Higher Education (HE) has become a focal point of discussion among tertiary practitioners, generating ongoing debate about its appropriate use (Zawacki-Richter et al., 2019). While AI's origins trace back to the 1950s, its presence in research emerged only three decades ago (Johnson and Lester, 2016; Norvig and Intelligence, 2002). Since 2018, AI's prominence in HE has grown, with

projections indicating continued expansion and potentially transformative implications for the sector (Cotton et al., 2024; Johnson and Lester, 2016). Among several AI Chatbots available to the public, ChatGPT, released in November 2022 by OpenAI in San Francisco, USA, has particularly challenged traditional HE approaches and caused apprehension. This advanced AI system can synthesise large amounts of internet-sourced information to generate responsive queries (AIOpen, 2022; Liebrezn et al., 2023; Lo, 2023).

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<https://doi.org/10.1016/j.nedt.2025.106796>

Received 23 December 2024; Received in revised form 16 April 2025; Accepted 2 June 2025

Available online 3 June 2025

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AI represents a remarkable technological advancement, offering numerous benefits to students, including efficient idea exploration and support with assessment preparation (Iskender, 2023; Liebrezn et al., 2023). Additionally, interacting with AI technology can improve students' technological proficiency, a highly valued employability skill (Iskender, 2023). AI has the potential to stimulate HE students' critical thinking skills through the creation of complex simulations in specific disciplines, such as nursing (Dave and Patel, 2023; Glauberman et al., 2023). As AI technologies evolve, these simulations are expected to become more sophisticated, allowing students to practice clinical skills and decision-making while engaging in real-world and captivating learning experiences (Glauberman et al., 2023; Hwang et al., 2024). Consequently, AI technologies in HE can facilitate the delivery of contemporary, technology-rich learning experiences, ultimately producing world-ready graduates who meet industry expectations (Glauberman et al., 2023).

AI already plays a vital role in hospital management, digital medical records, diagnostic guidance and imaging analysis in healthcare settings (Demir-Kaymak et al., 2024). Future nurses, midwives and other healthcare cadres are positioned to drive and shape the AI evolution in healthcare, necessitating effective preparation for AI-based technologies (Dave and Patel, 2023; Demir-Kaymak et al., 2024). As training institutions for the future healthcare workforce, HE must be at the forefront of understanding the challenges and opportunities posed by AI to reduce AI anxiety, improve AI readiness and awaken interest in technological innovations (Demir-Kaymak et al., 2024). Appropriate and ethical AI use can facilitate evidence-based practice implementation for health professionals, potentially narrowing the research-to-practice gap in healthcare (Fontenot, 2024).

While technology can enhance learning experiences, inappropriate AI use in assessments can compromise learning quality. Students who misuse AI in their assessments may face penalties for academic misconduct and fail to develop essential critical thinking skills, which can affect their learning outcomes and future workplace performance (Iskender, 2023). Cotton et al. (2024) advocate for developing resources to support ethical AI use, including clear policies and guidelines to clarify expectations for appropriate and ethical AI use in HE. This can prevent students and the university from experiencing 'sociotechnical blindness' (Hopcan et al., 2024). Sociotechnical blindness refers to the failure to identify or understand the ethical implications of using a particular technology, leading to a lack of accountability and transparency in its use within an organisation (Hopcan et al., 2024). Sociotechnical blindness can heighten students' anxieties and fears about the consequences of AI use (Almaiah et al., 2022; Hopcan et al., 2024; Lemay et al., 2020). While many higher education institutions have developed guidelines for AI use, some were knee-jerk reactions that did not consider students' perspectives and experiences with AI in their academic activities.

The first step to guiding the ethical and appropriate use of AI is to engage with stakeholders, including students, to understand their experiences and perceptions of AI use in their academic practices. However, there is limited and emerging knowledge regarding AI use and associated perceptions among nursing students. Previous research has primarily focused on acceptance and attitudes towards AI (Alsenany et al., 2024; Bhandari et al., 2021; Farghaly Abdelaliem et al., 2022) and intentions of AI use in healthcare services (Ahmad et al., 2023; Labrague et al., 2023). Limited attention has been given to understanding how nursing students integrate AI into their studies and the influence of their existing attitudes on its usage. Thus, this project aimed to investigate the extent to which nursing students integrate AI technologies into their studies and identify associated perceptions and demographics at an Australian university.

2. Methods

2.1. Design

This study employed a convergent parallel mixed methods approach to develop a comprehensive understanding of nursing students' experiences of AI use and associated perceptions in their academic work (Dawadi et al., 2021). Qualitative and quantitative data were collected concurrently to ensure that both types of data thoroughly captured and explained the phenomenon of AI use among nursing students (Dawadi et al., 2021).

2.2. Participants and setting

Convenience sampling was used to recruit undergraduate and postgraduate nursing students from a nursing school at a public university in Western Australia. Both undergraduate and postgraduate nursing students from all academic levels were invited to participate in the study.

2.3. Quantitative data measures

A survey was used to collect quantitative data. This survey included a question on AI use (dependent variable), in addition to eleven more questions on students' knowledge (dependent variable) and demographic data (dependent variable), including their age, gender, language used at home, academic level, being domestic or international students, were collected to understand the demographic characteristics of the sample. Further, a modified survey was used to collect data on perceptions (classified as independent variables). This modified survey was originally developed by Venkatesh et al. (2012) based on the unified theory of acceptance and use of technology (UTAUT) and later adapted by Strzelecki (2023). This survey was modified to fit the aim of this study by replacing the term 'ChatGPT' used in Strzelecki (2023) with 'AI'. A five-point Likert scale, with response options of strongly disagree, disagree, neutral, agree, strongly agree, was used across its seven domains of performance expectancy (4 items), effort expectancy (4 items), social influence (3 items), facilitating conditions (4 items), hedonic motivation (3 items), personal innovativeness (4 items), and habit behavioural intention (3 items). This tool has been used in Western cultures with a Cronbach's alpha of 0.82 and above across the seven domains (Strzelecki, 2023). The reliability of this tool in the analysis for this study ranged from a Cronbach's alpha of 0.82 to a Cronbach's alpha of 0.95 across its seven domains.

2.4. Recruitment and data collection

Following institutional approval, recruitment occurred through the school's learning management community sites. To maintain research integrity, a school's support officer, independent of the research team, distributed survey invitations to students containing both a QR code and a direct link to the survey. The Qualtrics survey employed a forced response functionality, requiring participants to confirm their informed consent before accessing the survey questions. Upon survey completion, participants were invited to participate in follow-up interviews.

The survey generated quantitative data on how students use AI in their written academic work. At the end of the survey, students were invited to volunteer for interviews. Interested students were asked to click on a link where they could leave their contact details, and a research assistant followed up to recruit students into the qualitative arm.

For the qualitative study, in-depth interviews with purposively sampled students were conducted by an experienced qualitative research assistant (EA1) who had no prior relationship with the participants, enhancing the study's credibility. An interview guide, comprising nine open-ended questions, was developed based on the current evidence on students' experiences of AI use in academia and

items from the adapted survey tool. Interviews were conducted either face-to-face, online via secured Microsoft Teams, or through telephone calls, depending on participants' preferences. All interviews were recorded and transcribed. Although audio-visual data was collected from some participants, only the audio data was transcribed verbatim for data analysis. After transcription, all video recordings were deleted from the password-protected computer to protect participants' privacy.

Qualitative data was collected until data saturation was achieved. Data saturation occurred with the tenth participant; however, three additional interviews were conducted to ensure no new themes emerged. In-depth interviews lasted between 30 and 40 min and were designed to extract data on the experience and use of AI, complementing the quantitative data. All data was collected between October 2023 and April 2024.

2.5. Data analysis

Quantitative data were analysed using SPSS version 28.0.1 (Corp., 2021). Descriptive statistics were applied to participant characteristics such as gender, age, educational level, employment status and use of AI. Descriptive statistics are presented as counts and percentages. Bivariate factors associated with AI use were assessed using the chi-square test. Factors that were significant at $p \leq 0.05$ in bivariate analysis were entered in multivariate logistic regression to identify predicting factors for AI use. The p -value was set at 0.05 for all analyses.

Qualitative data were analysed thematically following the verbatim transcription of interview data using NVivo 12. Guided by the principles of thematic analysis by Braun and Clarke (2019), the transcripts were independently coded by two researchers (MZ and EA2), who met periodically to agree on the codes. Based on similarity, MZ and EA2 independently grouped the coded data into subthemes, which formed themes. The coders maintained regular communication to discuss and agree on the emerging themes. Finally, the identified themes were verified by the three other researchers (JM, MC, and MB) before all authors carried out a critical analysis and interpretation of the derived themes. The iterative process of discussing and reflecting on the data and themes increased the rigour of the thematic analysis process.

2.6. Mixed methods analysis/Data triangulation

Following the in-depth analysis of both qualitative and quantitative data, data triangulation was undertaken using the convergent joint display approach. This method integrated and compared both data sets concerning the research question, providing a holistic view of students' experiences, perceptions and AI use in their studies (Guetterman et al., 2015). The integration of these two data sets enhanced the validity of the study findings. Where both methods reported similar findings, the consistency strengthened the credibility of our findings (Creswell and Plano Clark, 2018).

2.7. Ethical approval

Ethical approval was obtained from the university's Human Research Ethics Committee before the study commenced. Additionally, permission was secured from the University's Director of Research Services to access students. A study invitation email was sent to students via the learning management system. Participation in this study was contingent upon providing consent. A forced response was applied to the Qualtrics survey, requiring students to confirm they had read and consented to the study before accessing the survey. Participation was voluntary, and participants could stop or withdraw at any time.

3. Results

3.1. Quantitative findings

Overall, 188 undergraduate and postgraduate nursing students responded to the survey. However, this analysis is based on 105 valid responses. Most respondents were female (81.9 %). More than half of the participants (67.6 %) were studying on campus and 15 % had an inclusion and accessibility plan with the university. There were more domestic students (70.5 %) than international students (29.5 %), and a majority (63.8 %) spoke English at home. Regarding the level of studies, 85 % of the students were undergraduates, 20 % were completing postgraduate coursework or higher degree by research courses (see Table 1).

Data on students' perceptions of AI use showed that the majority (76.2 %) of nursing students disagreed with copying and pasting AI-generated text, felt the need to use AI (73 %), believed AI was becoming a habit (66.7 %), and were concerned about being addicted to AI use (75.2 %). However, almost half of the participants (41 %) found AI useful, and 41 % found it enjoyable (Table 2).

The results of the bivariate analysis of the data are presented in Table 3. The perception of the benefits of AI, the ease of AI use, the level of confidence in AI use, and the sources of AI information were significantly associated with the use of AI ($p \leq 0.05$). None of the socio-demographic characteristics of students were significantly associated with the use of AI ($p \geq 0.05$).

Table 4 shows factors associated with the use of AI among nursing

Table 1
Participants' characteristics.

Variables (n = 105)		Frequency	Percentage
Gender	Male	17	16.2
	Female	86	81.9
	Nonbinary	1	1
	Not disclosed	1	1
Mode of study	On campus	71	67.6
	Off-campus	34	32.4
Year of study	Year 1 or 2	82	39.4
	Year 3 or 4	23	26.0
	Other	3	34.6
Studied course	Undergraduate	85	81
	Postgraduate	20	19
Age group	Under 25	46	43.8
	25 to 29	14	13.3
	30 to 39	30	28.6
	Over 40	15	14.3
Inclusion and accessibility plan	No plan	82	78.1
	Has plan	15	14.3
	Prefer not to disclose	8	7.6
Employment status	Not employed	19	18.1
	Employed	86	81.9
Domestic or international student	Domestic	74	70.5
	International	31	29.5
Primary language at home	English	67	63.8
	Other languages	38	41.2
Used AI	A little to none	81	77.1
	Moderate/a lot	24	22.9
Heard about AI	Heard more	48	45.7
	A little to nothing	57	54.3
Heard from the news media	Yes	24	22.2
	No	84	77.8
Heard from social media	Yes	52	48.1
	No	56	51.9
Heard from other students	Yes	48	44.4
	No	60	55.6
Heard from the workplace	Yes	13	12
	No	95	88
Other sources	Yes	10	9.3
	No	98	90.7
Confidence in AI use	Not confident/neutral	81	77.1
	Confident	24	22.9

Table 2
Students' perceptions on AI.

Perceptions	Frequency	Percentage
I believe that AI is useful in my studies		
Disagree	19	18.1
Neutral	43	41.0
Agree	43	41.0
Using AI increases your chances of achieving important things in your studies		
Disagree	30	28.6
Neutral	33	31.4
Agree	42	40.0
Using AI helps you get tasks and projects done faster in your studies		
Disagree	18	17.1
Neutral	32	30.5
Agree	55	52.4
Learning how to use AI is easy for me		
Disagree	25	19.0
Neutral	46	40.0
Agree	34	41
My interaction with AI is clear and understandable		
Disagree	20	19.0
Neutral	42	43.8
Agree	43	37.1
I find AI easy to use		
Disagree	20	18.1
Neutral	46	35.2
Agree	39	46.7
AI increases your productivity in your studies		
Disagree	19	23.8
Neutral	37	43.8
Agree	49	32.4
I have the knowledge necessary to use AI		
Disagree	31	29.5
Neutral	28	26.7
Agree	42	40.0
Using AI is fun		
Disagree	12	11.4
Neutral	50	47.6
Agree	38	36.2
Using AI is enjoyable		
Disagree	12	11.4
Neutral	45	42.9
Agree	43	41.0
Using AI is very entertaining		
Disagree	14	13.3
Neutral	52	49.5
Agree	34	32.4
Using AI has become a habit for me		
Disagree	70	66.7
Neutral	17	16.2
Agree	13	12.4
I must use AI		
Disagree	73	73
Neutral	20	20
Agree	7	73
People who are important to me think I should use AI		
Disagree	35	33.3
Neutral	42	40.0
Agree	25	23.8
I have the knowledge necessary to use AI		
Disagree	40	38.1
Neutral	41	39.0
Agree	21	20.0
People whose opinions I value prefer me to use AI		
Disagree	41	39.0
Neutral	39	37.1
Agree	22	21.0
I am addicted to using AI		
Disagree	79	75.2
Neutral	14	13.3
Agree	7	6.7
Using AI has become natural for me		
Disagree	65	61.9
Neutral	24	22.9
Agree	11	10.5

Table 2 (continued)

Perceptions	Frequency	Percentage
I will always try to use AI in my studies		
Disagree	46	43.8
Neutral	41	39.0
Agree	12	11.4
I plan to continue to use AI frequently		
Disagree	42	40.0
Neutral	40	38.1
Agree	17	16.2
Using AI is a form of academic cheating or plagiarism		
Disagree	15	14.3
Neutral	31	29.5
Agree	43	41.0
I am familiar with the need to reference the use of AI in my work		
Disagree	16	15.2
Neutral	17	16.2
Agree	56	53.3
I copy and paste exactly what Generative AI develops		
Disagree	80	76.2
Neutral	6	5.7
Agree	3	2.9
I intend to continue using AI in the		
Disagree	25	23.8
Neutral	33	31.4
Disagree	25	23.8

students. Students who were more informed about AI were four times more likely to use it than those who had heard little to nothing about AI (aOR 3.9; 95 % CI: 1.07–10.2). Additionally, students who perceived AI as useful in their studies were nearly six times more likely to use it than those who did not (aOR 5.5; 95 % CI: 1.7–17.3). Regarding perceptions, students who found using AI easy were 3 times likely to use AI than their counterparts (aOR 3.4; 95 % CI: 1.1–11.1).

4. Qualitative results

Analysis of the qualitative data resulted in three themes and seven sub-themes. Details of the themes are explained below:

4.1. Theme 1: integration of artificial intelligence in studies

This theme shows how participants apply AI to their usual academic tasks in HE. It is made up of three sub-themes, as discussed below.

4.1.1. The study coach and editor

Only two students reported having less knowledge about AI, and three respondents reported not using AI. Those aware of AI technology used it at various levels for different purposes. For example, some students used AI as their academic coach or tutor, incorporating it into their academic writing and assignment requirements. Some students only recently “*found out about ChatGPT*” and “*...started using it in their assignments*” (Participant 12). Some students used AI as a paraphrasing tool when writing assignments. Students felt “*secure*” using AI to double-check ideas and ask questions. This is exemplified in the following quote by Participant 7, “*I’m working on my proposal, and I use ChatGPT to check things out. I always have the ChatGPT window open while writing*” (Participant 7). Participants shared their ongoing challenges when completing assignments and how the emergence of AI has helped to address their writing needs. For many who identified English as an additional language (EAL), paraphrasing ideas from academic sources was “*the most challenging part of the assignments*” as they had “*limited vocabularies*. Additionally, some students found “*putting the assignment together*” as the most challenging part of writing their academic essays. However, with the use of AI, they found these academic challenges including “*starting an assignment, constructing its structure and gathering information*” easier.

Participants whose first language is English and those from culturally

Table 3
Factors associated with the use of AI in bivariate analysis.

Variable	Has not used AI N (%)	Has used AI N(%)	p-Value
Whether a domestic or an international student (<i>n</i> = 105)			0.329
Domestic	59 (56.2)	15 (14.3)	
International	22 (21)	9 (8.6)	
Age (<i>n</i> = 105)			0.810
< 25 years	36 (34.3)	10 (9.5)	
> 25 years	45 (42.9)	14 (13.3)	
Campus location (<i>n</i> = 105)			0.542
On campus	56 (53.3)	15 (14.3)	
Off-campus	25 (23.8)	9 (8.6)	
Year of study (<i>n</i> = 105)			
Year 1–2	52 (49.5)	15 (14.3)	
Year3–4/Other	29 (27.6)	9 (8.6)	
The language used at home (<i>n</i> = 105)			0.879
English	52 (49.5)	15 (14.3)	
Language other than English	29 (27.6)	9 (8.6)	
Employment status (<i>n</i> = 105)			0.157
Not employed	17 (16.2)	2 (1.9)	
Employed	64 (61)	22 (21)	
Heard about AI? (<i>n</i> = 105)			0.001**
A little to nothing	6 (5.7)	18 (17.1)	
Heard more	30 (28.6)	26 (46.4)	
Heard from the news (<i>n</i> = 105)			0.776
Yes	18 (17.1)	6 (5.7)	
No	63 (60)	18 (17.1)	
Heard from social media (<i>n</i> = 105)			0.465
Yes	37 (35.2)	13 (12.4)	
No	44 (41.9)	11 (10.5)	
Heard from work (<i>n</i> = 105)			0.468
Yes	9 (8.6)	4 (3.8)	
No	72 (68.6)	20 (19)	
Other students (<i>n</i> = 105)			0.291
Yes	34 (32.4)	13 (12.4)	
No	47 (44.8)	11 (10.5)	
Other sources (<i>n</i> = 105)			0.821
Yes	8 (7.6)	2 (1.9)	
No	73 (69.5)	22 (21)	
How confident are you? (<i>n</i> = 105)			0.012**
Not confident	67 (63.8)	14 (13.3)	
Confident/mixed feelings	14 (13.3)	10 (9.5)	
I believe that AI is useful in my studies (<i>n</i> = 105)			<0.001***
Disagree/Neutral	58 (55.2)	23 (21.9)	
Agree	4 (3.8)	20 (19)	
AI increases chances of achieving important things in your studies (<i>n</i> = 105)			0.002**
Disagree/Neutral	55 (52.4)	8 (7.6)	
Agree	26 (24.8)	16 (15.2)	
AI use helps you get tasks and projects done faster in your studies (<i>n</i> = 105)			0.012*
Disagree/Neutral	44 (41.9)	6 (5.7)	
Agree	37 (35.2)	18 (17.1)	
AI increases productivity in your studies (<i>n</i> = 105)			0.402
Disagree/Neutral	45 (42.9)	11 (10.5)	
Agree	36 (34.3)	13 (12.4)	
Learning how to use AI is easy for me			<0.001***
Disagree/Neutral	58 (55.2)	4 (3.8)	
Agree	23 (21.9)	20 (19)	
My interaction with AI is clear and understandable (<i>n</i> = 105)			<0.001***
Disagree/Neutral	60 (57.1)	6 (5.7)	
Agree	21 (20)	32 (17.1)	
I find AI easy to use (<i>n</i> = 105)			<0.001***
Disagree/Neutral	52 (49.5)	4 (3.8)	
Agree	29 (27.6)	20 (19)	
People who are important to me think I should AI (<i>n</i> = 102)			0.842
Disagree/Neutral	60 (58.8)	17 (16.7)	
Agree	19 (18.6)	6 (5.9)	

Table 3 (continued)

Variable	Has not used AI N (%)	Has used AI N(%)	p-Value
People who influence my behaviour believe that I should use AI (<i>n</i> = 102)			0.056
Disagree/Neutral	66 (35.4)	15 (14.7)	
Agree	13 (12.7)	8 (7.8)	
People whose opinions I value prefer I use AI (<i>n</i> = 102)			0.549
Disagree/Neutral	63 (61.8)	17 (16.7)	
Agree	16 (15.7)	6 (5.9)	
I have necessary knowledge to use AI (<i>n</i> = 101)			0.002**
Disagree/Neutral	52 (51.5)	7 (6.9)	
Agree	26 (25.7)	16 (15.8)	
AI use is fun (<i>n</i> = 100)			<0.001***
Disagree/Neutral	55 (55)	7 (7)	
Agree	23 (23)	15 (15)	
AI use is enjoyable (<i>n</i> = 100)			0.001***
Disagree/Neutral	51 (51)	27 (27)	
Agree	6 (6)	16 (16)	
AI use is very entertaining (<i>n</i> = 100)			0.005**
Disagree/Neutral	57 (57)	9 (9)	
Agree	21 (21)	13 (13)	
The use of AI has become a habit for me (<i>n</i> = 100)			0.024*
Disagree/Neutral	71 (71)	16 (16)	
Agree	7 (7)	6 (6)	
I am addicted to using AI (<i>n</i> = 100)			0.020*
Disagree/Neutral	75 (75)	20 (20)	
Agree	5 (5)	2 (2)	
I must use AI (<i>n</i> = 100)			0.663
Disagree/Neutral	73 (73)	20 (20)	
Agree	5 (5)	2 (2)	
AI use has become natural for me (<i>n</i> = 100)			<0.001***
Disagree/Neutral	75 (73)	14 (14)	
Agree	3 (3)	8 (8)	
I intend to continue using AI in the future (<i>n</i> = 99)			0.007**
Disagree/Neutral	68 (68.7)	14 (14.1)	
Agree	9 (9.1)	8 (8.1)	
I will always try to use AI in my studies (<i>n</i> = 99)			0.014*
Disagree/Neutral	71 (71.1)	16 (16.2)	
Agree	16 (16.2)	6 (6.1)	
AI use is a form of academic cheating or plagiarism (<i>n</i> = 89)			0.100
Disagree/Neutral	33 (37.7)	13 (14.6)	
Agree	37 (41.6)	6 (6.7)	
I am familiar with the need to reference the use of AI in my work (<i>n</i> = 89)			0.981
Disagree/Neutral	26 (37.1)	7 (7.9)	
Agree	44 (49.4)	12 (13.5)	

*** $p < 0.001$.

** $p < 0.01$.

* $p < 0.05$.

and linguistically diverse (CALD) backgrounds noted that AI enhances their written academic English language skills. For instance, Participant 7 mentioned, “90% of the time, the problem is with my English writing” (Participant 7). Other participants shared that without AI, they “can’t find synonyms for some words”, leading them to “repeating some words numerous times”. As a result, they “ask ChatGPT for alternative words and structure” to improve the quality of their written assignments. The use of AI for editorial purposes is exemplified in the following quote, “I like to use it for editing purposes because I struggle a lot with academic writing. I’ll be writing on something for so long, and I struggle to judge how readable it is to another person” (Participant 2).

4.1.2. AI use for researching information

Participants shared how they use AI to search for information, clarify concepts, and seek “advice”, “feedback”, and “guidance” when

Table 4
Factors associated with the use of AI in multivariate logistic regression analysis.

Variable	aOR (95 % CI)
I believe AI is useful in my studies	
Disagree/Neutral	Reference
Agree	5.5 (1.7–17.3)**
AI increases my chances of achieving important things in my studies	
Disagree/Neutral	Reference
Agree	1.2 (0.4–4.0)
Learning how to use vAI is easy for me	
Disagree/Neutral	Reference
Agree	3.4 (1.1–11.1)*
How confident are you in using AI?	
Not confident	Reference
Confident/mixed feelings	0.9 (0.3–3.0)
Heard about AI	
A little to nothing	Reference
Heard more	3.9 (1.07–10.2)*
AI use helps you get tasks and projects done faster in your studies	
Disagree/Neutral	Reference
Agree	0.351 (0.124–1.0)
I will always try to use AI in my studies	
Disagree/Neutral	Reference
Agree	1.2 (0.43–3.6)
I intend to continue using AI in the future	
Disagree/Neutral	Reference
Agree	2.7 (0.7–10)
AI use has become natural for me	
Disagree/Neutral	Reference
agree	1.8 (0.56–5.6)
My interaction with AI is clear and understandable	
Disagree/Neutral	Reference
Agree	1.7 (0.3–9.6)
I know necessary to use AI	
Disagree/Neutral	Reference
Agree	1.1 (0.41–2.78)
AI use is fun	
Disagree/Neutral	Reference
Agree	1.4 (0.159–13.3)
AI use is enjoyable	
Disagree/Neutral	Reference
Agree	1.4 (0.18–11.1)
AI use is very entertaining	
Disagree/Neutral	Reference
Agree	0.93 (2–1.4)
The use of AI has become a habit for me	
Disagree/Neutral	Reference
Agree	0.75 (0.18–3.2)
I am addicted to using AI	
Disagree/Neutral	Reference
Agree	1.2 (0.3–5)

* $p < 0.05$; aOR adjusted for all variables in the table.

** $p < 0.01$.

undertaking academic work or projects. Some participants even used AI to search “for references” to include in their academic work. For example, a master’s student shared that, “It [AI] can also give guidance on referencing, which is what I used it on, but then it’s not quite confident in that regard” (Participant 10). Most participants used AI to research academic concepts that were difficult to comprehend or to seek feedback or advice on their written work. AI was used to seek answers to questions such as “What is a mixed-method study?” (Participant 11).

4.2. Theme 2: motivational and deterrent factors of AI use

This theme highlights the contextual factors influencing students’ use of AI. The extent to which students engage with AI for academic purposes is influenced by the nature of AI and the students’ internalised and externalised values.

4.2.1. The appeal of AI capabilities

Most students found AI attractive due to its capabilities, which represented a paradigm shift in HE. Describing AI as “intelligent” and “fascinating”, many students were excited by its ability to provide synthesised or rectified information “perfectly” and in a “timely” manner. This was perceived as an effective and efficient alternative to time-consuming information-searching strategies, such as using Google or reading a book or article, as evidenced by this quote:

“You ask a question, and it answers straight away. It tailors the answer because you feed it what you need, and then it finds the information for you. Sometimes, it gives you a link, and I use the link to see if that’s good enough to use”

(Participant 8)

4.2.2. Disincentivising AI use

The use of AI was often linked to individual students’ internalised values. These values were anchored in finding satisfaction in academic work that reflected their true academic abilities. This was common among students who believed in themselves and wanted to increase their knowledge and academic writing skills. As stated by Participant 5, “When I do my assignment myself, the feedback that I get helps me go through the areas that I need to work on. It’s not just copy-pasting what AI has done and then submitting it”. These students leveraged their weaknesses to improve their academic abilities and believed that AI use would deter growth, thus, they opted out.

On the other hand, students with self-doubt in their academic abilities were more inclined to use AI. For instance, one student said, “I have become very dependent on it [ChatGPT], and I don’t trust myself anymore.... I’ve been using ChatGPT to check almost everything. Even when I want to write a birthday card for someone, it’s like I have no ideas for myself anymore. So, for anything that requires a lot of thinking, I have to ask ChatGPT. And it’s the same thing in my research. The worst part is that I can’t come up with original ideas by myself anymore” (Participant 7).

This perception of dependency on AI in some students’ academic work extended to concerns about their hindered ability to think independently and frustration over the lack of original and independent thinking.

Participants described their negative experiences with AI, specifically ChatGPT. Students described ChatGPT as “not always correct” and “inaccurate”, which was deemed unsafe for nursing practice. Participant 10 narrated, “I discovered ChatGPT through a colleague at work who is a pain specialist nurse. We were discussing a patient’s pain relief, and she was asking me for advice. She said she would put these in ChatGPT and see what it recommends. And yes, it was inaccurate”. Such students had an aversion to AI or used it with caution.

4.2.3. Ethical concerns

The study revealed multifaceted ethical concerns among nursing students regarding AI use, including issues of academic integrity, professional development, and future practice implications. These concerns manifested through four primary dimensions: plagiarism, anxiety, learning quality, and professional preparation. Participants shared worries and fears about AI use for academic purposes, linking these negative feelings to their values, past experiences, and understanding of academic integrity. Students expressed significant anxiety about potential breaches of academic integrity, particularly regarding the uniqueness of AI-generated content. This was illustrated by a participant who feared plagiarism due to AI potentially generating identical content for multiple students:

I’m scared that I’ll get done for plagiarism. How do you not know it’s not giving that to somebody else? Everyone in the class will have the same assignment, so if I tell AI to write me an introduction about renal stones, someone in the class says that, too. How do you not

know that other people in your class will not have the same AI output as you?

(Participant 4)

I presume that if I present perfect assignments, the lecturer might have questions like how are you doing it like this? Then they may say, your work is so good; please tell us your understanding of it, or I might be asked to be a group leader and teach my peers. Yet you don't have the information to actually do that because it was not your work

(Participant 6)

...students [AI use] will not learn the trade [nursing] and will be less prepared for practice. And the fact that AI will become better with time is also worrying. It will be faster, and maybe students will be able to write a whole assignment in a minute, and their teacher might not even notice it. Who knows?

(Participant 12)

Similarly, some participants feared using AI because its intelligence surpasses the average student for a respective study level or stage. Such students feared submitting perfect assignments that would draw their lecturer's attention. Students articulated concerns about becoming "more lazy", "not progressing in my writing skills", and "not learning" and, therefore, being "less prepared for practice", reflecting a deeper worry about the fundamental nature of education and skill acquisition. The evolving sophistication of AI technology has created additional ethical dilemmas for students, as "the teacher might not notice" an AI written assignment in the future.

4.3. Theme 3: yearning for support and information

This theme shows the need for targeted information and support for students as they engage with AI in their academic activities. The extent to which they can use AI and still achieve learning outcomes remains unclear, causing misunderstandings and concerns for students. It consists of two sub-themes: unclear guidelines and the need to feel challenged.

4.3.1. Unclear guidelines

Students expressed their need for AI education and guidance, including 'clear' information and training on how to use AI appropriately in their academic work to maintain academic integrity. Participants wanted the university to provide criteria and consequences for AI use, as exemplified by Participant 8: "I don't know what I can use it or can't use it for.... it would be good if the information came from uni [university] to show us how to use it safely. Uni needs to tell us about the criteria for AI use, which, if not adhered to, then there are consequences. That needs to be done and explained clearly." This lack of clarity in the use of AI induced fear and anxiety among students, which was further heightened by observable uncertainty from academic staff, as demonstrated in this excerpt: "When my supervisor said that it's not okay to use AI in my research, she wasn't sure, but she said that they might check our browsers to see if we use ChatGPT. After that, I got so scared about using it" (Participant 7). When asked about the preferred method of education on the use of AI, participants mentioned a "step-by-step video", "tutorial", or "workshop" and "resources." Students also suggested including more support and feedback for those who use AI tools.

4.3.2. The need to feel challenged while embracing AI

Participants perceived AI as valuable when they hit a roadblock in understanding, allowing them to seek assistance and then delve deeper into the topic through additional research and learning, as highlighted in this quote: "I think when students run out of ideas, they can ask AI and then do further research and learning to deepen their understanding of the topic" (Participant 11). They adopted a pragmatic stance, recognising AI's inevitable integration into education and advocating for a balanced

approach from educational institutions. Instead of implementing outright bans, participants recommended that institutions articulate their position on AI usage, accompanied by well-reasoned justifications and clarifications. Participants commented that the rejection of AI "seems harsh", given the prevalence of technology even in basic writing tools like Word documents. They compared AI's role to basic writing tools like spelling and grammar correction, emphasising the need to incorporate AI into academic practices... "AI will become a part of our lives and studies. When they say don't use AI at all, it seems harsh because even Word documents give us some type of grammar or spelling corrections. Uni should clarify and justify the reasons for or against" (Participant 12).

While most students found AI to be "inevitable" and had a "positive attitude towards embracing AI use for studies", some participants expressed the need to be challenged to ensure they could not rely solely on AI to complete their assessments. One participant suggested "... maybe include more quizzes and have oral exams in class" (Participant 12), whilst another proposed the use of AI "... to make the process of studying easier and not make AI do 100% of our work" (Participant 10). Although AI can assist with studying, the participants asserted that they needed to learn and understand information themselves to apply it correctly and avoid knowledge gaps in their future careers as nurses and midwives. One participant concluded, "... let them [AI applications] make the work easier, but don't do my job fully. At least give me my part to challenge myself".

5. Discussion

As AI becomes more popular, students are increasingly incorporating it into their studies. The key finding from this study is that university students use AI for various study tasks, influenced by their level of AI awareness and both internalised and externalised values. This discussion combines qualitative and quantitative data on two levels of AI awareness and use, as well as students' perceptions of AI use. Additionally, the interplay between demographic characteristics and the use of AI is explored.

5.1. AI awareness and use

Aligning with other research reports, the majority of participants in both qualitative and quantitative arms had heard about AI (Labrague et al., 2023; Singh et al., 2023; Stöhr et al., 2024), and increased exposure to AI technologies optimised the likelihood of its usage. These findings are consistent with extant literature, where knowledge and awareness of AI were identified as facilitators of AI adoption (Labrague et al., 2023; Stöhr et al., 2024). This suggests that increasing awareness of AI's advantages and appropriate use could enhance students' engagement with the technology. HE institutions should provide ongoing AI training to students, considering the evolving nature of AI to promote its safe adoption. This approach will help students to stay current and adapt to technological advancements crucial to nursing practice. Training methods should also consider students' preferred ways of receiving information, such as social media and peer students, as these factors also predicted engagement with AI technologies in this study.

Approximately one-quarter of nursing students who completed the survey reported AI use, a figure notably lower than previous studies, which reported 69 % and 46 % rates in Johnston et al. (2024) and Chang et al. (2024), respectively. In contrast, 10 out of 13 participants in the qualitative arm reported AI use. While AI use was not categorically accepted by participants, more than half of the survey respondents agreed that "using AI helps you get tasks and projects done faster in your studies", and about 41 % found AI enjoyable. It is acknowledged here that the low reported usage rate may underestimate the actual AI engagement by this population. This disparity between reported usage and perceived benefits suggests a possibility of underreporting, a phenomenon also noted in another study (Johnston et al., 2024). The timing

of data collection for this study coincided with the initial emergence of AI chatbots in HE (2023) when institutional policies were still forming and unclear, which may have influenced students to provide more socially acceptable responses in the survey.

Aligning with previous research reports (Johnston et al., 2024), additional data from qualitative interviews show that students use AI as a study coach, assisting with editorial work, explaining complex concepts, and quickly finding information. Baïdoo-Anu and Owusu Ansah (2023) regard AI use as a virtual tutor as an ethical approach to integrating AI in teaching and learning activities, enhancing students' understanding and motivation for self-directed learning (Zhou and Li, 2023).

5.2. Students' perceptions and AI use

In this study, several perceptions were significantly associated with AI usage in quantitative analysis, with the belief in *the usefulness of AI* being the only predictor for its use. The impetus for using AI technologies often stems from an individual's attraction to the technology and personal values (Zhou and Li, 2023). Similar to previous studies (Stöhr et al., 2024; Zhou et al., 2024), students in both arms of this study were drawn to use AI because of its *appealing capabilities*, which participants found entertaining and useful for their studies. However, the attractive features of AI technologies present a potential risk of over-reliance among students. Evidence of addictive characteristics was observed in both arms of the study. As future nurses, over-reliance on AI undermines the development of critical thinking skills and the application of clinical judgement, which depend on human interpretation and touch to make sound clinical decisions and implement interventions (Nashwan and Abujaber, 2023; Ronquillo et al., 2021). Hua (2023) contends that while an overdependence on technology may not directly compromise academic integrity, it becomes problematic when students rely on these tools to complete assessments. Such dependence can adversely affect the development of students' critical thinking and writing skills. George and Wooden (2023) emphasise the need to balance AI innovation and ethical practice, prompting the need to carefully examine what constitutes ethical practice around the use of AI in HE.

Building on the findings of Chen et al. (2024), more than half of students were unsure or did not perceive AI usage as a form of academic misconduct. Interestingly, a higher percentage of students reported refraining from directly copying and pasting AI-generated content. Conversely, within the theme of *Ethical Concerns* in qualitative analysis, many students cited fears of violating academic integrity and not acquiring essential skills for future practice as reasons for their discomfort with AI use, which supports previous findings (Singh et al., 2023; Stöhr et al., 2024). This may explain why only 22.9 % reported confidence in AI use, aligning with findings from previous studies (Pedlow and Maldon, 2024; Pedlow and Maldon, 2023). Increased exposure could improve student confidence levels, aligning with findings from a study by Kelly et al. (2023), which demonstrated a positive correlation between experience and confidence in AI use ($p = 0.001$, $\rho = 0.494$). Furthermore, confidence in using technology increases as digital literacy improves (Australian Institute for Teaching and School Leadership, 2024; Stöhr et al., 2024). The inability or lack of interest in using technology impacts confidence and can cause stress and anxiety, which is concerning in today's fast-paced technological society (Zhou and Li, 2023).

These integrated findings suggest an underlying tension between technological advancement and maintaining academic authenticity, which has implications for universities. This anxiety about AI's impact on learning authenticity connects to broader concerns about professional competency development, suggesting that students recognise the crucial link between authentic academic work and future clinical practice capabilities. These findings have implications for universities to provide clear guidelines on how and when students can use AI, and the consequences of inappropriate AI use should be enforced. The data also

highlighted the need for readily accessible AI resources, ideally co-designed with students. Higher education stakeholders should develop clear policies to support students' use of AI, as it has become integral to their learning and assessment preparation and will continue to be so.

Restricting students from using technologies could be a backwards step, potentially disadvantaging students from AI learning support (Baïdoo-Anu and Owusu Ansah, 2023; Johnston et al., 2024). Currently, the use of AI is banned in some universities and units/courses within training programs, and some institutions use AI detection tools in assessment; however, it is still unclear whether these strategies enhance academic integrity and professional competence (Sullivan et al., 2023). By embracing AI with a nuanced understanding of its capabilities and limitations, educational institutions can empower students to navigate the digital landscape effectively and increase their confidence while simultaneously preparing them for the evolving technological demands of the future. Providing clear guidance through educational interventions on appropriate AI use will help students to feel comfortable with the technology, increasing the opportunity to engage with it, and it will also enhance their learning outcomes, including the development of technological skills (Singh et al., 2023; Zhou and Li, 2023). Looking ahead, longitudinal studies with larger, multi-institutional samples are needed to track the evolution of AI use and its impact on learning outcomes, specifically on professional competency, academic performance and critical thinking skills. This inquiry should extend to the integration of AI in clinical settings and its impact on the quality of care and dependency.

As AI technology continues to evolve, educational institutions must work proactively to create frameworks that support ethical and effective AI use while preparing nursing students for an increasingly technology-enhanced healthcare environment. The need to incorporate scaffolded learning opportunities about the ethical use of AI in the curricula cannot be overstated. These should clarify the implications of over-reliance on AI, the expected level of interaction with AI technologies, acknowledgement and referencing, and the abilities and shortfalls of AI. In addition, HE should redesign authentic learning activities and assessments and promote the direct application of critical thinking skills by designing tasks that enable the students to articulate their perspectives. This could comprise designing assessment tasks, learning activities that incorporate specific AI prompts, critiquing AI-generated responses to specific prompts, submission of annotated drafts and investing in employing high-skilled staff to facilitate students' education on responsible use of AI to maintain academic integrity. Sullivan et al. (2023) suggest 'podcasts, oral presentations, laboratory activities, group work, handwritten work, participation grades, vivas, and very specific assignment prompts' as examples of authentic assessments. Further, there is a need to clarify ethical or appropriate AI use in HE for a common language and consistency essential for policy development (Fitzpatrick and McCarthy, 2016).

5.3. Demographic data and AI use

Almost all participants in the qualitative arm used AI. However, the quantitative analysis did not establish a significant correlation between AI use and any of the demographic variables in the present study. A previous study on Australian university students with a larger sample size found similar results (Kelly et al., 2023). Interestingly, being an international student ($p > 0.5$), using English as an additional language ($p > 0.5$) and being a domestic student ($p > 0.5$) did not have any statistically significant impact on AI use. Qualitative findings augment this finding, where students who believed in themselves when completing assignments averted AI use. Singh et al. (2023) conducted a study on university students in the UK and found that most students used AI for language-based tasks. However, less than half of the international students used AI to improve their English language skills. On the other hand, scholars in China used AI to enhance their academic writing skills (Li et al., 2024). Further research with meticulous methodologies and

larger sample sizes is needed to resolve this conflict. However, it is argued here that both internalised and externalised factors influence the ethical or unethical use of AI.

5.4. Strengths and limitations

This study demonstrates several significant strengths that enhance its contribution to understanding AI use in nursing education. The convergent parallel mixed-methods approach provided complementary data that offered both breadth and depth in understanding nursing students' AI use. The quantitative data established clear patterns and associations, and the qualitative insights illuminated students' underlying reasoning and experiences, creating a comprehensive understanding of the phenomenon. The study benefited from a diverse participant pool that included both undergraduate and postgraduate students, on-campus and off-campus learners, domestic and international students, and those from various cultural and linguistic backgrounds. This diversity in the sample enhanced the representativeness of the findings and provided insights across different student demographics.

Furthermore, the timing of data collection during the initial phase of AI integration in HE captured early adoption patterns and concerns, establishing valuable baseline data for future comparative studies. Therefore, this study contributes to the growing body of knowledge about AI in HE and provides a foundation for future research into the effective integration of AI in nursing education. The findings can inform policy development and educational practices as institutions navigate the opportunities and challenges presented by AI technology in academic settings.

It should be noted that the data used in this study was collected between October 2023 and April 2024, offering a time-bound snapshot of students' experiences and use of the evolving AI technology. It is expected that the reported findings may not represent the current usage and awareness. Furthermore, the convenient sample for this study was taken from one public Australian university, which may limit the generalisation of findings to other industries and settings.

The application of the convergent mixed methods approach strengthened the findings of this study, whereby the limitation of one method was mitigated by the other. For example, quantitative data provided a broad overview of trends, frequencies, and patterns in the use and perceptions of AI in nursing education, while the qualitative method delved deeper into specific areas or elements of interest. For example, the survey reported that 7 % of participants were addicted to AI use without explaining 'the conditions considered in the addiction'. On the other hand, data from the interviews provided the depth and context of AI addiction to enhance the complete understanding of the phenomenon (Creswell and Plano Clark, 2018).

6. Conclusion

This study provides valuable insights into the complex landscape of AI use among nursing students in HE. The findings suggest that successful integration of AI in nursing education requires a nuanced approach that acknowledges students' ethical concerns, provides clear institutional guidance, maintains focus on professional competency development, and leverages AI's potential while preserving authentic learning experiences. HE should consider redesigning learning activities and assessments that are authentic, reflect real-world situations and require the direct application of critical thinking and practical problem-solving skills. In addition, educational institutions need to develop transparent, well-justified guidelines for AI use that balance technological innovation with academic integrity. There is a pressing need for structured support systems that address both technical skills and ethical considerations in AI use. Furthermore, the integration of AI education should consider the unique context of nursing education, where clinical competence and patient care are paramount.

CRedit authorship contribution statement

Maggie Zgambo: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Martina Costello:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Melanie Buhmann:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Justine Maldon:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Edah Anyango:** Writing – review & editing, Investigation, Data curation. **Esther Adama:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

There is no financial, personal or academic conflict of interest.

Acknowledgements

We would like to thank all nursing students who participated in the study.

Note

The oral paper was presented at the 9th International Nurse Education Conference. (NETNEP 2024, Singapore, 27–30th October).

Grants and financial support

This study was supported by the university's School of Nursing and Midwifery Executive Dean's Award – 2023 Teaching and Learning Grant.

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